FPGA Neural Network - Final Report

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Project: Iris Flower Classifier (4 -> 8 -> 3)
Target Hardware: Basys 3 FPGA (Artix-7)

1. Introduction

The objective of this project was to design, train, and implement a 2-layer artificial neural network on an FPGA to classify the Iris flower dataset. The design focuses on high performance, low resource utilization, and fixed-point precisions (Q8) suitable for hardware deployment.

2. Architecture & Design

The system follows a sequential feed-forward architecture:
- Input Layer: 4 normalized features.
- Hidden Layer: 8 neurons utilizing the ReLU activation function.
- Output Layer: 3 neurons utilizing an Argmax approach for prediction.

Hardware Implementation:
- neuron.v: Implements a MAC unit with bias addition and ReLU clipping.
- layer.v: Parametrized module to chain multiple neurons.
- nn_top.v: Orchestrates the data flow (FSM) between layers and memory.

Figure 1: FPGA Neural Network Hardware Architecture

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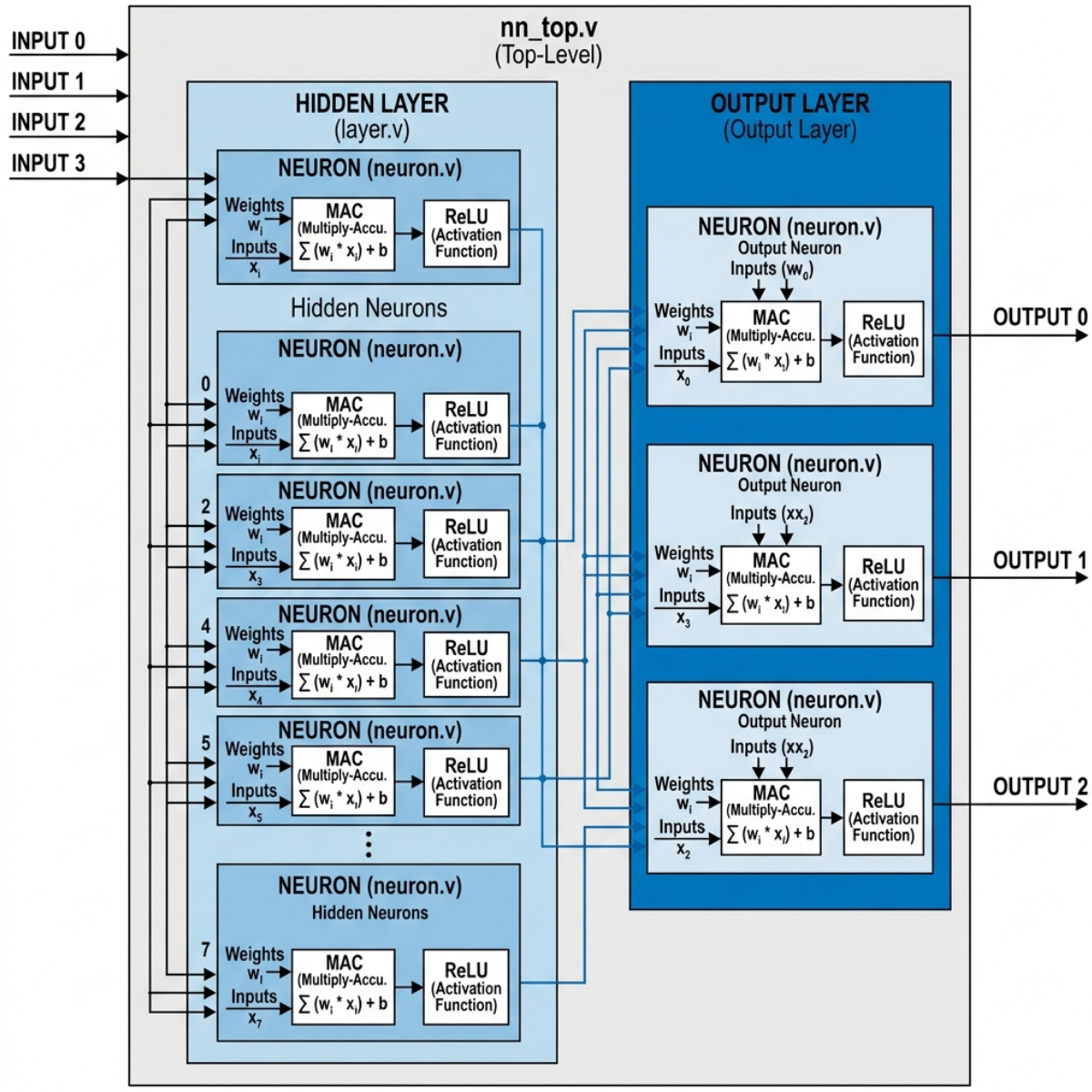

3. Training & Software Results

The model was trained using TensorFlow/Keras and exported to 16-bit signed Q8 fixed-point format.

- Test Accuracy: 93.3%
- Max Absolute Weight: 1.1219 (Well within Q8 range - no overflow).

4. Verification Results

Both simulation and hardware synthesis confirmed the logic.

Sample ID	Expected Class	Predicted Class	Status
0	0 (Setosa)	0	PASS
1	2 (Virginica)	2	PASS
2	1 (Versicolour)	1	PASS

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3	1 (Versicolour)	1	PASS
4	0 (Setosa)	0	PASS

Note: Simulation was verified against 10 test samples from test_data.mem.

5. FPGA Performance Metrics (Vivado)

Metric	Value
Logic Utilization (LUTs)	14 (<1%)
Registers	15 (<1%)
Worst Negative Slack (WNS)	6.866 ns
Operating Frequency	100 MHz (Target)

6. Conclusion

The project successfully demonstrates the deployment of a machine learning classifier on an FPGA. The resource footprint is extremely small, making it suitable for larger-scale integration or edge-device applications.

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